5 Experimental Evaluation

	ACC (%)				UR (%)				SR (%)			
	SE	GK	SA	AVG	SE	GK	SA	AVG	SE	GK	SA	AVG
Llama-3-8B	80.5	16.1	68.0	54.9	**68.9**	39.0	60.3	**56.1**	**10.2**	14.7	16.2	**13.7**
Mistral-7B	77.6	11.4	64.0	51.0	56.3	22.5	59.1	46.0	3.6	**17.9**	14.1	11.9
Zephyr-7B-beta	70.9	9.9	65.8	48.9	38.9	13.8	63.2	38.6	8.1	12.3	13.4	11.3
Qwen2.5-7B	**85.0**	**24.8**	**74.2**	**61.3**	56.0	**38.5**	**67.9**	54.1	2.1	4.0	4.9	3.7
DeepSeek-R1-7B	84.1	22.5	56.2	54.3	54.8	28.3	43.4	42.2	5.2	9.1	**18.9**	11.1

Table 1: Token-level perturbation results on SingleEQ (SE), GSM8K (GK), and StrategyQA (SA) datasets. We report the accuracy (ACC), unattackable rate (UR), and success rate (SR) for five models across three datasets. The results reveal that the models have a significant sensitivity to the token-level perturbations.

	ACC (%)				UR (%)				SR (%)			
	SE	GK	SA	AVG	SE	GK	SA	AVG	SE	GK	SA	AVG
Llama-3-8B	80.5	16.1	68.0	54.9	24.9	41.8	21.2	29.3	**5.1**	10.8	29.7	**15.2**
Mistral-7B	77.6	11.4	64.0	51.0	**88.3**	**63.6**	**79.7**	**77.2**	0.0	**18.5**	2.8	7.1
Zephyr-7B-beta	70.9	9.9	65.8	48.9	70.6	54.6	77.2	67.5	2.2	13.1	2.1	5.8
Qwen2.5-7B	**85.0**	**24.8**	**74.2**	**61.3**	1.9	46.8	18.3	22.3	0.5	1.8	24.0	8.8
DeepSeek-R1-7B	84.1	22.5	56.2	54.3	20.6	59.3	10.0	30.0	6.1	6.4	**33.1**	**15.2**

Table 2: Embedding-level perturbation results on SingleEQ (SE), GSM8K (GK), and StrategyQA (SQ) datasets. We report the accuracy (ACC), unattackable rate (UR), and success rate (SR) for five models across three datasets. The results reveal that the models have a significant sensitivity to the input perturbations, like embedding-level perturbations.

5.1 Experimental setup

Datasets. We use three datasets in line with existing literature on reasoning (Wei et al., 2022b; Trivedi et al., 2022; Kojima et al., 2022b; Diao et al., 2023). For math reasoning, we select two math datasets SingleEq (Koncel-Kedziorski et al., 2016) with single-step math problems and GSM8K (Cobbe et al., 2021) with multi-step math problems. Apart from that, we select StrategyQA (Geva et al., 2021) for commonsense reasoning. For all datasets, we use the test split for the evaluation. Specifically, we sampled 500 questions for the evaluation of the StrategyQA dataset and used all data (508 for SingleEq and 1319 for GSM8K) in the other two datasets. More details about the datasets can be found in Appendix A.2.

Models. Our experiments focus on open-source and closed-source models for robustness evaluation. For open-source models, we use Llama-3-8B (Dubey et al., 2024a), Mistral-7B (Jiang et al., 2023), Zephyr-7B-beta (Tunstall et al., 2023), Qwen2.5-7B (Yang et al., 2024), and DeepSeek-R1-Distill-Qwen-7B (Guo et al., 2025b). We evaluate the transferability of white-box pertrubations to closed-source models GPT-4 and GPT-3.5-turbo (Achiam et al., 2023). Also, we use GPT-3.5-turbo as our judging model for comparing COTs, and DeepSeek-R1-Distill-Qwen-7B (Guo et al., 2025b) for evaluating the semantic meanings of perturbed questions. Human evaluations of those judges are available in Appendix A.4 and Appendix A.7.

Metrics. For all datasets, we report the accuracy (ACC), attack success rate (SR), and unattackable rate (UR). ACC represents the percentage of correctly answered questions without perturbations; SR represents the percentage of problems answered with wrong reasoning when the answers are correct after the perturbation; UR indicates the percentage of problems answered correctly with correct reasoning steps when the answers are correct after the perturbation. WR=1-UR-ASR represents the percentage of questions answered correctly before but answered with wrong answers after the perturbation, which is common in our settings. Also, UR is used to represent the robustness of reasoning perturbations (e.g., a higher UR means a higher robustness).

Implementation. For token-level perturbation on open-source models, we set the inserted ratio to be $a = 0.2$ for all datasets; for the replacement ratio k, we set it to be $(0.5, 0.25, 0.5)$ for SingleEq, GSM8K, and StrategyQA datasets, respectively. For embedding-level perturbation, we choose hyperparameters with perturbation percentage $\epsilon = (0.02, 0.005, 0.03)$ for SingleEq, GSM8K, and StrategyQA datasets, which are imperceptible for the token mapping. Also, we set the number of perturbation steps j to be 5 for all datasets

and two kinds of perturbations. For closed-source model experiments, we set temperature $T = 0.7$ and we test the closed-source models on the adversarial examples that open-source models generate. The default version of GPT-3.5-turbo used is GPT-3.5-turbo-0125, and the GPT-4o used is GPT-4o-2024-08-06. Apart from that, we set the maximum length of generating output to be 256. We use a conventional ICL setting for our experiments (Brown et al., 2020; Rae et al., 2021). We apply the same number of exemplars as Wei et al. (2022b) and use the exemplars from Diao et al. (2023), which are 8 for SingleEq and GSM8K, as well as 6 for StrategyQA. Runtime, LLM judge analysis, and limitations can be found in Appendix A.5 and A.7.

5.2 Comparison with baselines

In Table 3, we compare MATCHA with the following baselines for token-level perturbations: (1) **Random**: we randomly insert tokens into the original question; (2) **GCG**: we optimize a suffix similar to Zou et al. (2023) using our defined loss; (3,4,5) **MATCHA**: we experiment with MATCHA with random locations/with only CoT loss/full versions. We show that MATCHA can achieve a comparatively lower unattackable rate and higher success rate, indicating MATCHA token-level perturbations have stronger perturbation effectiveness.

5.3 Evaluation of open-source models

In Table 2 and Table 1, we show that MATCHA successfully examines and reveals the sensitivity of different models against embedding-level and token-level perturbations across three datasets (two math datasets and one commonsense reasoning dataset). We analyze the performance of our evaluation framework as follows.

	UR	SR
Random	49.1(0.7)	10.0(0.6)
GCG	65.4(7.6)	12.7(6.2)
MATCHA (random locs)	**36.0**(7.5)	12.2(5.0)
MATCHA (only $\mathcal{L}_c$)	41.5(2.9)	13.0(2.4)
MATCHA	39.0(5.0)	**14.7**(2.4)

Table 3: Baseline comparisons on GSM8K dataset using Llama-3-8B.

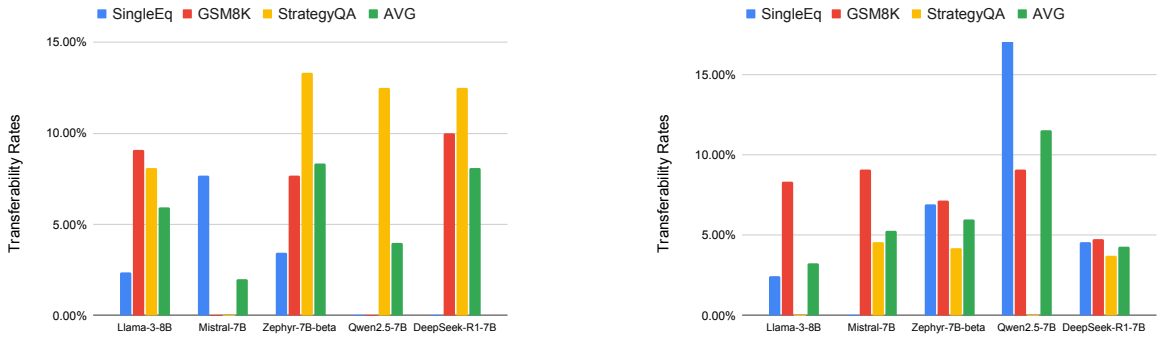
Embedding-level v.s. token-level attacks. The relative strength of perturbations at the embedding-level versus the token-level varies between models. For Mistral-7B and Zephyr-7B-beta, token-level perturbations yield lower UR, indicating stronger degradation of reasoning robustness. In contrast, for Llama-3-8B, Qwen2.5-7B, and DeepSeek-R1-7B, embedding-level perturbations are more effective, leading to greater reductions in UR. SR tends to be comparable or slightly higher for embedding-level perturbations, yet the non-trivial SR indicates the effectiveness of MATCHA. We observe that two kinds of perturbations reveal different aspects of model vulnerabilities with different behaviors across models.

Comparison of models. Mistral-7B and Zephyr-7B-beta show stronger robustness to embedding-level perturbations, with high UR, but are more vulnerable to token-level perturbations, suggesting their reasoning may be more sensitive to discrete token changes. In contrast, Llama-3-8B, Qwen2.5-7B, and DeepSeek-R1-7B are more affected by embedding-level perturbations, exhibiting greater drops in UR, which indicates a greater sensitivity to fine-grained perturbations in the input space. Further, Llama-3-8B and DeepSeek-R1-7 have the relatively highest SR, displaying the most severe inconsistency between reasoning and answer; Qwen2.5-7B has few inconsistent behaviors.

Comparison of datasets. Models exhibit lower robustness on multi-step reasoning (GSM8K) compared to single-step questions (SingleEq), likely because MATCHA can disrupt intermediate steps, leading to complete failure. Since multi-step reasoning depends on sequential logic, small perturbations can have a cascading effect, making these tasks more vulnerable. Models show moderate/low robustness on commonsense reasoning (StrategyQA), suggesting implicit knowledge is more vulnerable to perturbations than structured reasoning. While models handle logical reasoning, they remain susceptible to subtle perturbations in knowledge-based inference. Overall, multi-step and commonsense reasoning is more fragile under MATCHA, and improving both structured and knowledge-based answer/reasoning consistency is essential.

ACC v.s. UR. The accuracy-UR trade-off is more pronounced under embedding-level perturbations: high ACC models like Qwen2.5-7B and DeepSeek-R1-7 exhibit significantly lower UR, while lower-accuracy models such as Mistral-7B and Zephyr-7B-beta are more resilient. This suggests that accuracy-optimized models may develop sharper decision boundaries or more fragile internal representations, making them more vulnerable to fine-grained embedding perturbations. In contrast, token-level peturbations show a less consistent pattern, possibly because discrete substitutions interact with model tokenization in architecture-specific ways, leading

to irregular or less predictable robustness degradation. These results highlight the importance of evaluating both perturbation types when analyzing the robustness of LLM reasoning.



(a) Transferability to GPT-3.5-turbo.

(b) Transferability to GPT-4o.

Figure 2: Transferability to closed-source models (GPT-3.5-turbo and GPT-4o) experiments using the token-level successful examples, showing non-trivial transfer rates to the open-source models.

5.4 Transferability to closed-source models

Figure 2 illustrates the transferability rates of our token-level perturbations to closed-source models GPT-3.5-turbo and GPT-4o. The results indicate non-trivial transferability, suggesting that successful perturbations generated for open-source models can still effectively fool black-box models.

Model-level analysis: Qwen2.5-7B produces notably high transferability to GPT-4o, especially on structured tasks like SingleEq and GSM8K, indicating that its perturbations align closely with GPT-4o's decision boundaries. Zephyr-7B and Mistral-7B show strong transferability to GPT-3.5-turbo on different tasks, suggesting that base model capabilities and reasoning biases shape adversarial generalization.

Dataset-level analysis: Tasks like StrategyQA exhibit high transferability to GPT-3.5-turbo but much lower transferability to GPT-4o, suggesting that commonsense reasoning attacks transfer less effectively to more advanced models like GPT-4o. In contrast, GSM8K shows more consistent transferability, likely due to its focus on multi-step math reasoning, which may yield more structurally generalizable perturbations.

These results underscore that both types of reasoning in the dataset and the characteristics of the base model jointly determine transferability.

5.5 Examples of successful attacks

We categorize the successful attack examples into the following four types of errors and select four examples (one per category) across different models/datasets in Figure 3.

Wrong calculations refers to the errors in the calculation steps. As shown in the first row of Figure 3, the model thinks "20/3=5", which is incorrect in the calculation. **Wrong reasoning** refers to the evident reasoning errors in the procedure of arriving at the correct solution. For the second example in Figure 3, the model makes the mistake of getting the total number of running hours (should be $3 + 2/3 * 2 = 6$ instead of 4.5), leading to the overall error in the reasoning procedures. **Wrong information** refers to the model outputting some information that is wrong according to the question, to arrive at the correct answer. We present an example in the third column of Figure 3, where the model makes up "Sophocles" instead of "Sophist" to arrive at the correct answer. **Unrelated information** refers to the model outputting some unrelated information when trying to answer the question. In the fourth column of Figure 3, the model starts to make up non-existent information like "There are 15 trees in the grove" to guess the answer. The complete list of successful attack examples can be found in Appendix A.6.